Room Model

Software Requirements / Logical Data Model

1. Purpose

Represents a room within a hostel and its accommodation characteristics.

2. Responsibilities

• Identify the room and its parent hostel.

• Provide a structure for tracking room capacity and bed inventory.

• Support room availability and administrative room management.

3. Proposed Logical Attributes

The following is a documentation proposal, not a claim that every field is already implemented. Confirm exact field names, types, constraints and cardinalities against the project ERD/database schema.

Attribute	Meaning / Requirement
room_id	Unique room identifier.
hostel_id	Reference to the hostel containing the room.
room_number_or_code	Room label within the hostel.
floor_or_location	Optional location details.
capacity	Maximum configured occupants or beds, according to the system's definition.
status	Availability/maintenance state if supported.

4. Relationships

• Each room should belong to a hostel.

• A room may contain multiple beds.

• Room details may be used by allotment and availability reporting.

5. Validation and Business Rules

• Room number/code should be unique within its hostel unless the university defines a different rule.

• Configured capacity must be positive and consistent with bed records.

• A room should not be allocated beyond its supported capacity.

• Room deactivation or maintenance should not silently erase historical allotments.

6. Traceability / Notes

Confirm whether capacity is stored on Room, inferred from Bed records, or both; avoid conflicting sources of truth.

Status: Draft for review. Align this document with the approved SRS, ERD, class diagram, and institutional policies before treating it as final.